



WHAT IS DATA SCIENCE?

THE COMPLETE JOURNEY FROM DATA TO DECISION



DATA

UNDERSTAND

PREDICT

DECIDE

ACT

IMPACT



TURNING DATA INTO UNDERSTANDING.
UNDERSTANDING INTO ACTION. ACTION INTO IMPACT.



PRESENTED BY

DAKSH KHADELWAL
FOUNDER, THINKHUB



LEARN



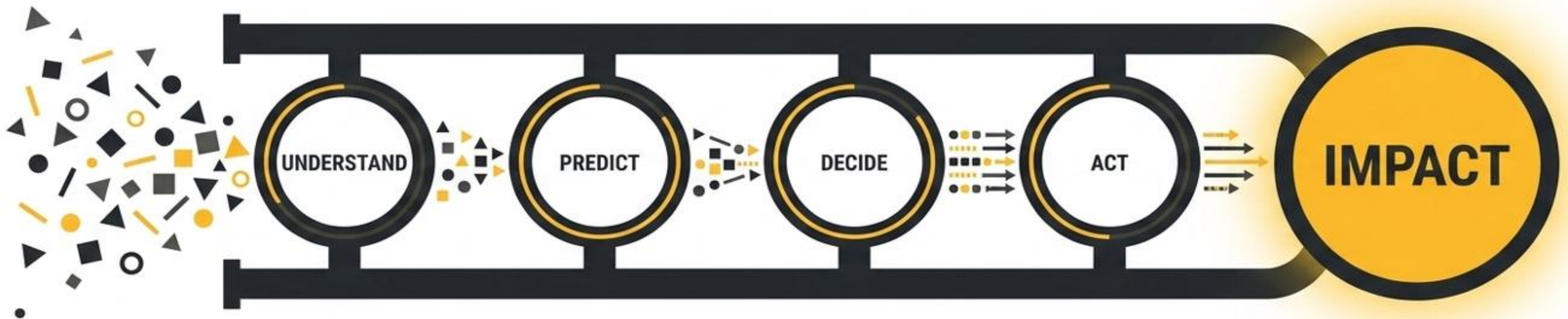
GROW



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What Is Data Science?

Turning data into evidence for better decisions.



How raw numbers become real-world decisions



Illustrative Example: E-Commerce Conversion

Data isn't the same as insight

04 — INSIGHT :

Meaning that influences choice
(e.g., Promotion drives sales)

03 — KNOWLEDGE :

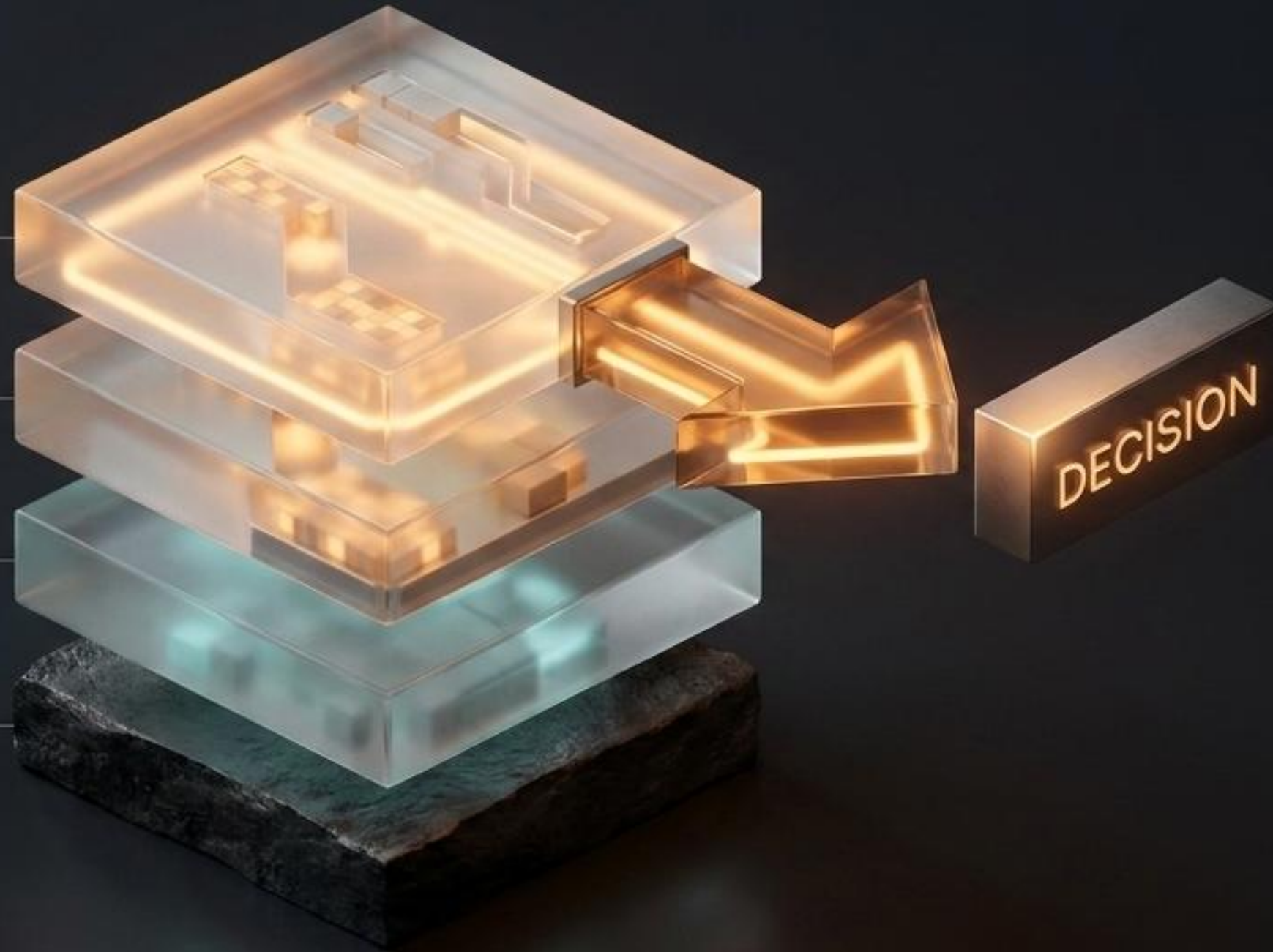
Recognized pattern
(e.g., Sales increase when promoted)

02 — INFORMATION :

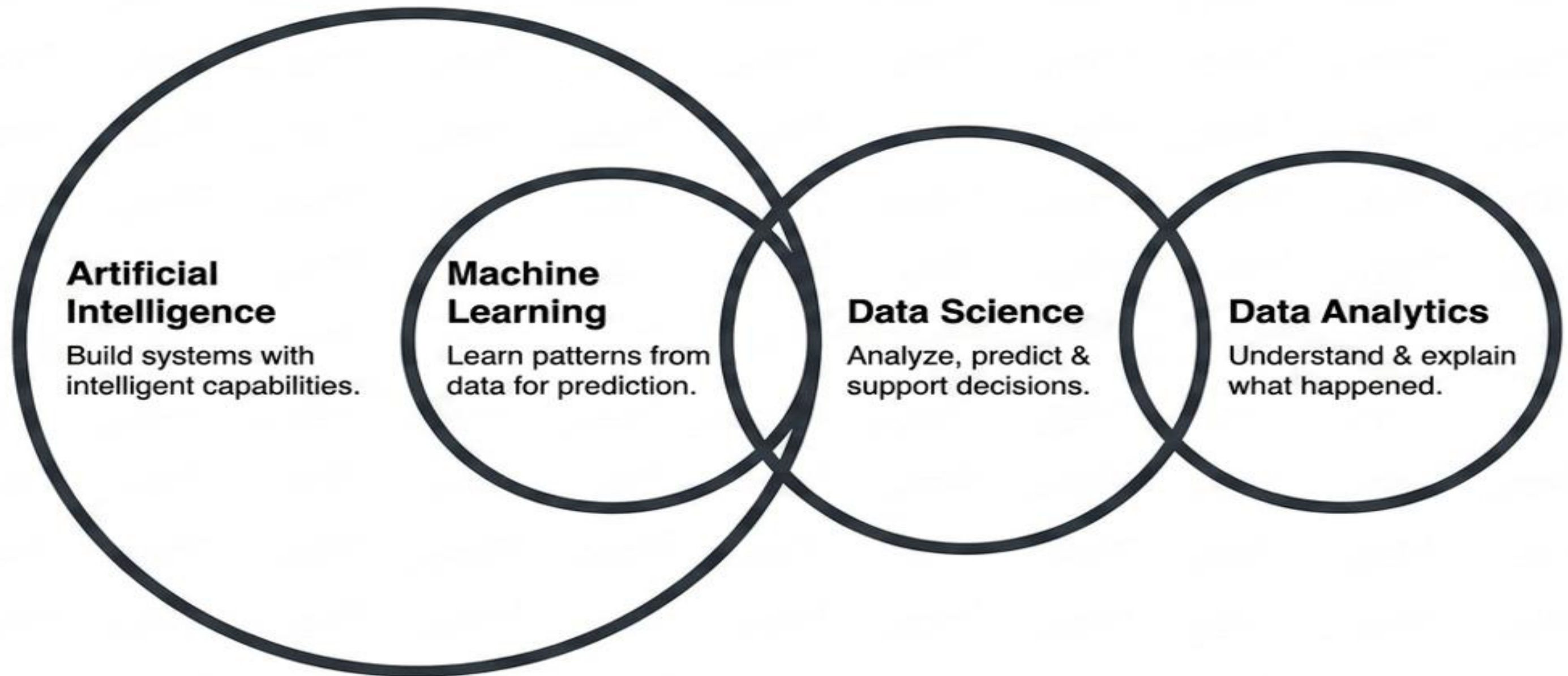
Contextualized data
(e.g., Product sold for ₹1,499)

01 — DATA :

Raw observations
(e.g., ₹1,499)



Mapping the ecosystem

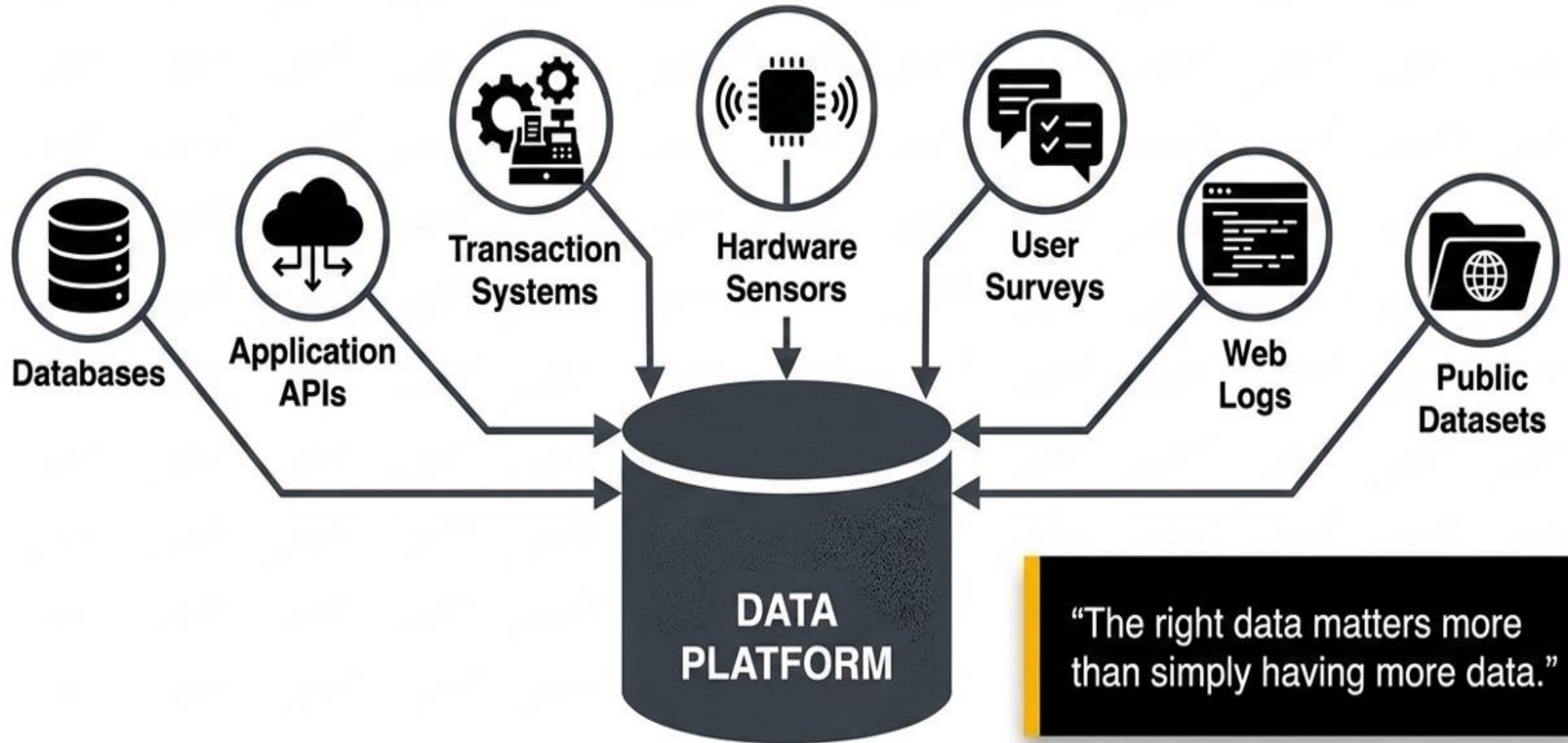


Machine Learning is a tool. Data Science is a broader problem-solving discipline.

The real Data Science lifecycle



Before analysis: Where does data come from?



"The right data matters more than simply having more data."

The data we get is rarely ready

ID	Name	Age	City	Date
1	Name: John Doe	Age: 32	New York	2022-10-03
2	Name: Parks	Age: -4	New York	2022-02-23
3	Name: Tolsey	Age: 1	Unknown	2022-03-33
4	Name: Typo	Age: 45	City: NULL !	2022-09-12
5	Name: Cochson	Age: 45	City: New York	
6	Name: Typos	Age: 150	City: New York	
7	Name: Typos	Age: 32	City: New York	
8	Name: John Doe	Age: 32	City: New York	
9	Name: John Doe	Age: 32	City: New York	
10	Name: John Doe	Age: 32	City: New York	
11	Name: Tim	Age: 3	City: New York	2022-10-13
12	Name: Chris	Typos -4	City: NULL	

Handle Missing Values
Remove Duplicates
Standardize Formats
Address Outliers

ID	Name	Age	City	Date
1	Name: John Doe	32	Unknown	2022-10-02
2	Name: Parks	45	New York	2022-03-28
3	Name: Tolsey	28	Unknown	2022-09-03
4	Name: Typo	32	Unknown	2022-09-02
5	Name: Cochson	45	New York	2022-10-02
6	Name: Typos	28	Unknown	2022-09-03
7	Name: Typos	33	Unknown	2022-09-07
8	Name: John Doe	32	Unknown	2022-09-26
9	Name: Rlay	28	Unknown	2022-10-07
10	Name: Slarie	55	Unknown	2022-10-03
11	Name: Sink	55	Unknown	2022-10-18
12	Name: Felick	28	mean value	2022-10-23

Garbage in → Garbage out.

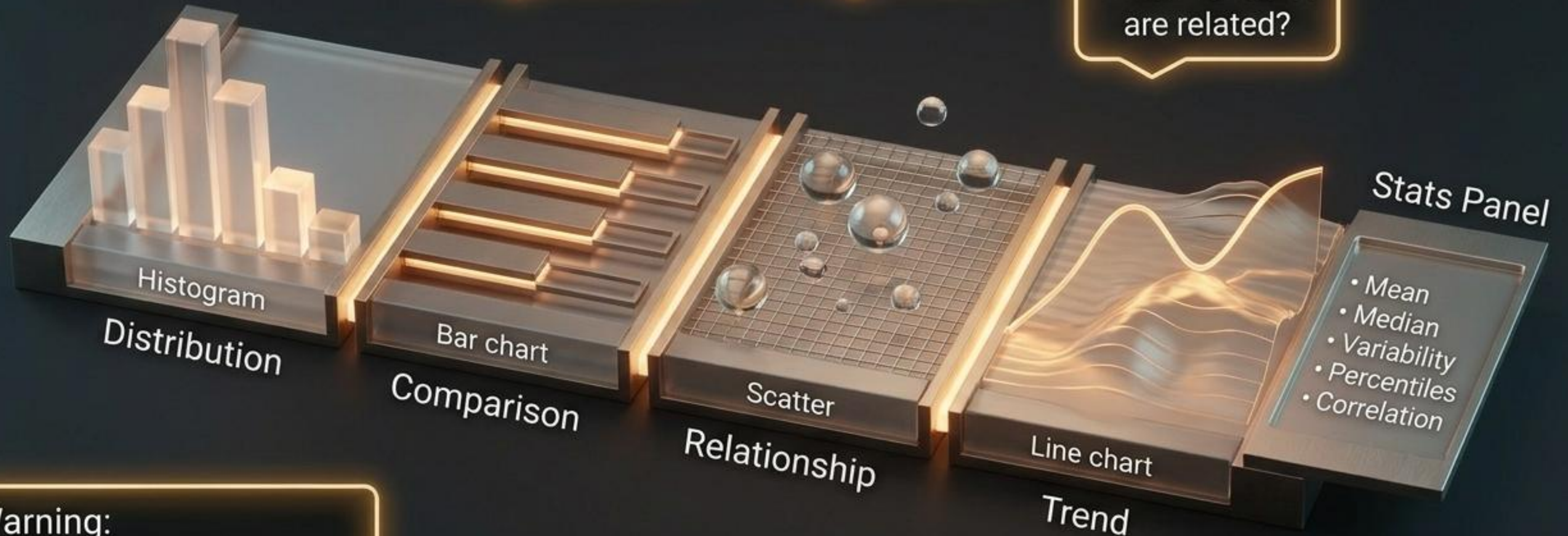
Before building a model, understand the data

What happened?

What patterns exist?

What looks unusual?

Which variables are related?



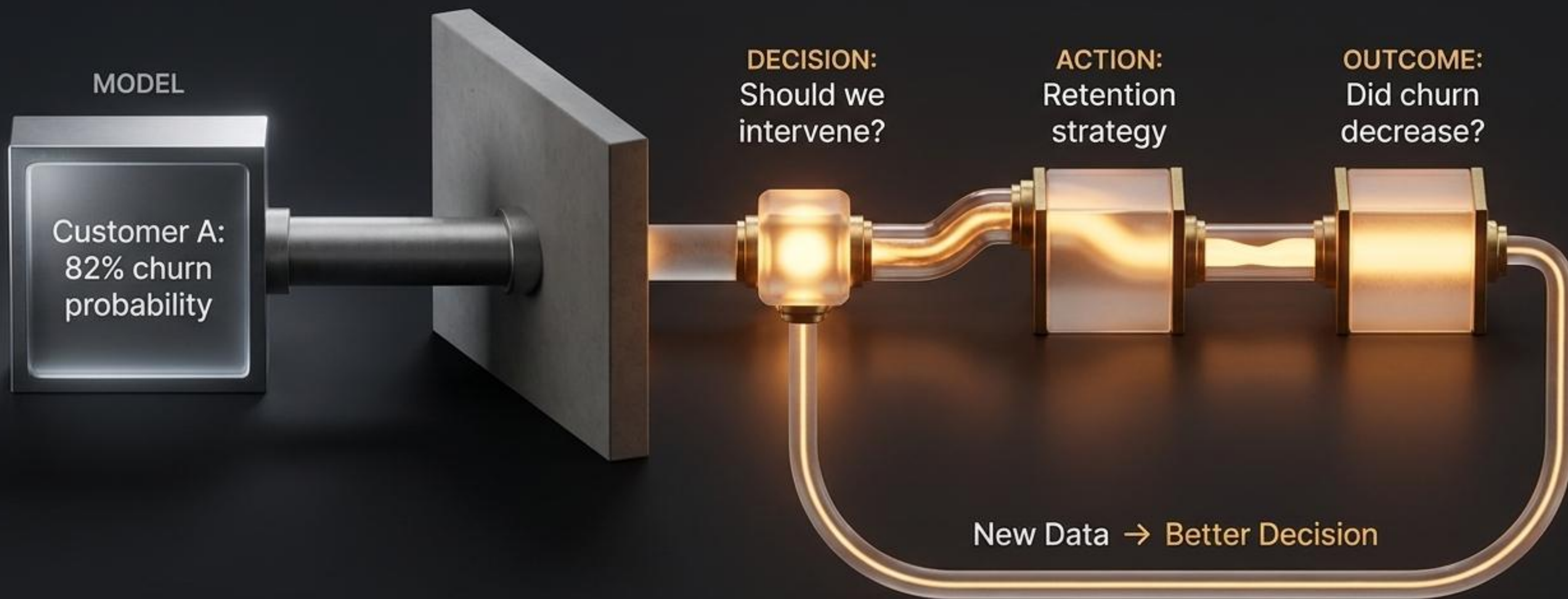
Warning:
Correlation $\neq$ Causation

When prediction becomes useful



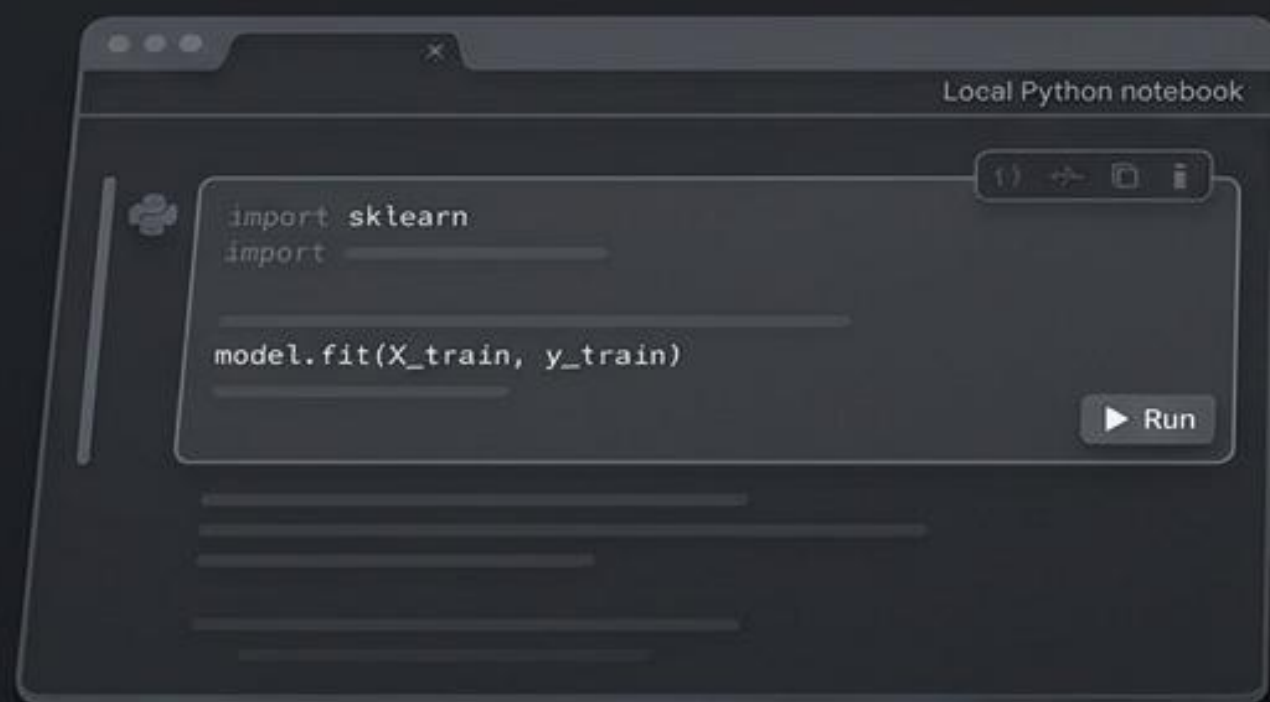
Not every **Data Science** problem needs **Machine Learning**.

A prediction is **not** a decision

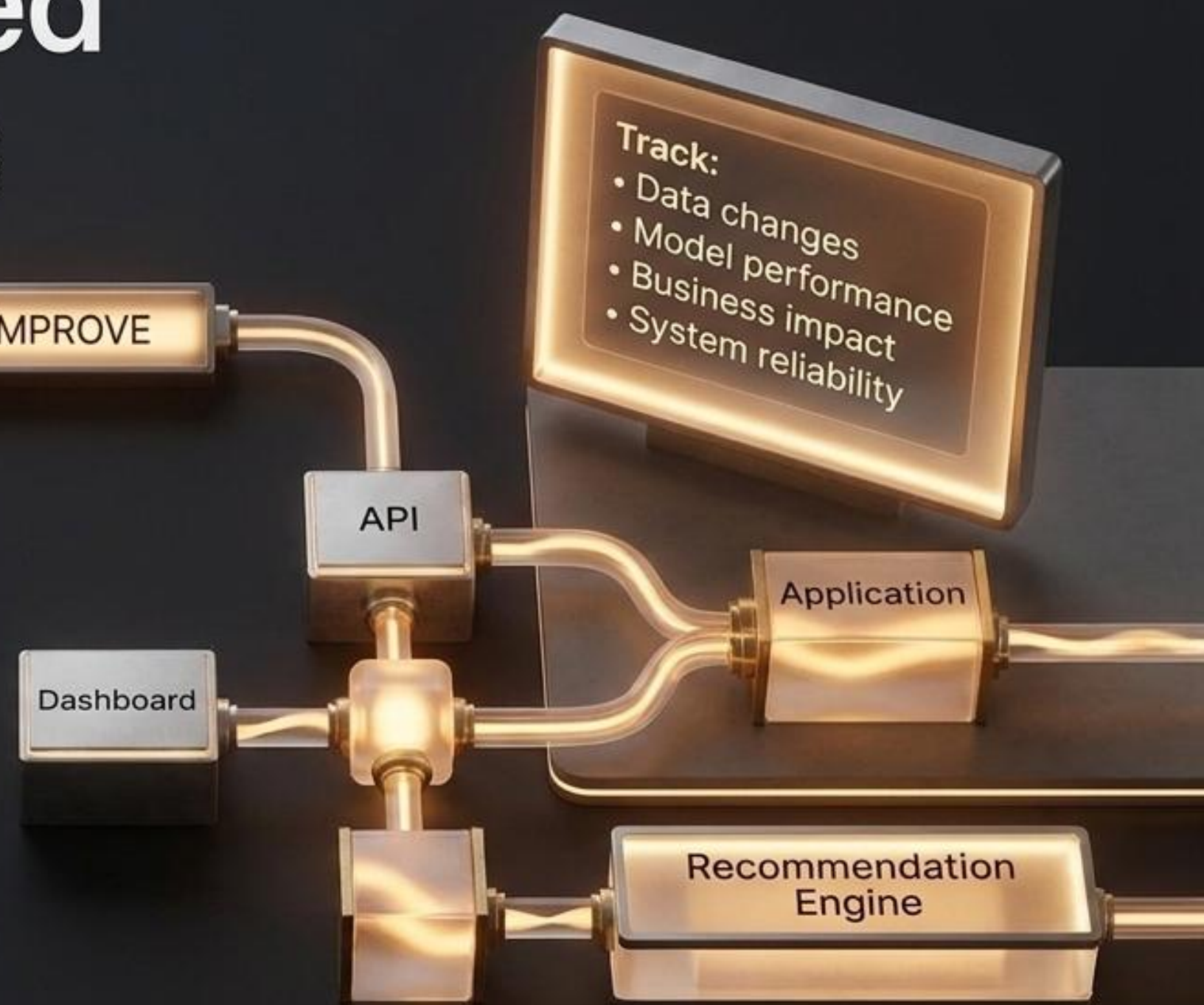


A model isn't finished when the code runs

TRAIN → EVALUATE → DEPLOY → MONITOR → RETRAIN / IMPROVE



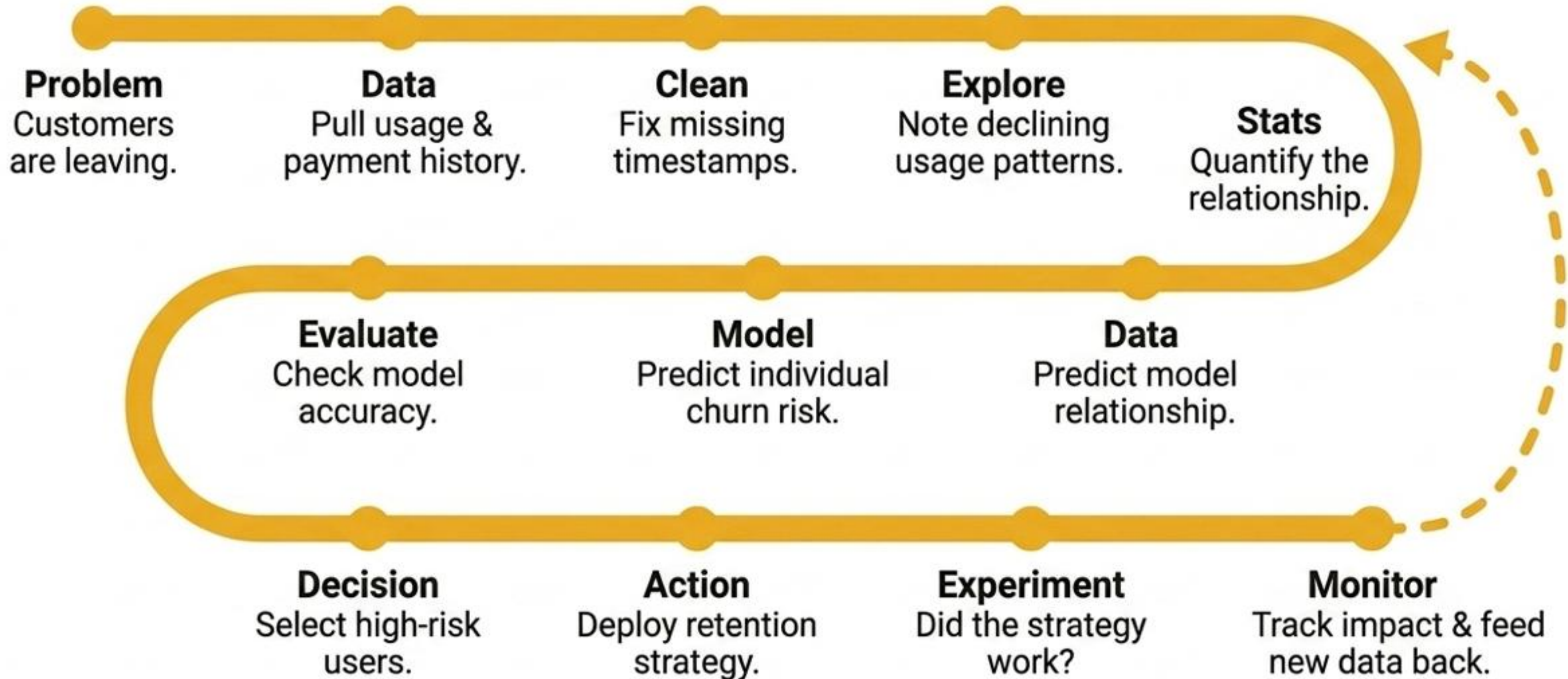
Left Side (The Lab)



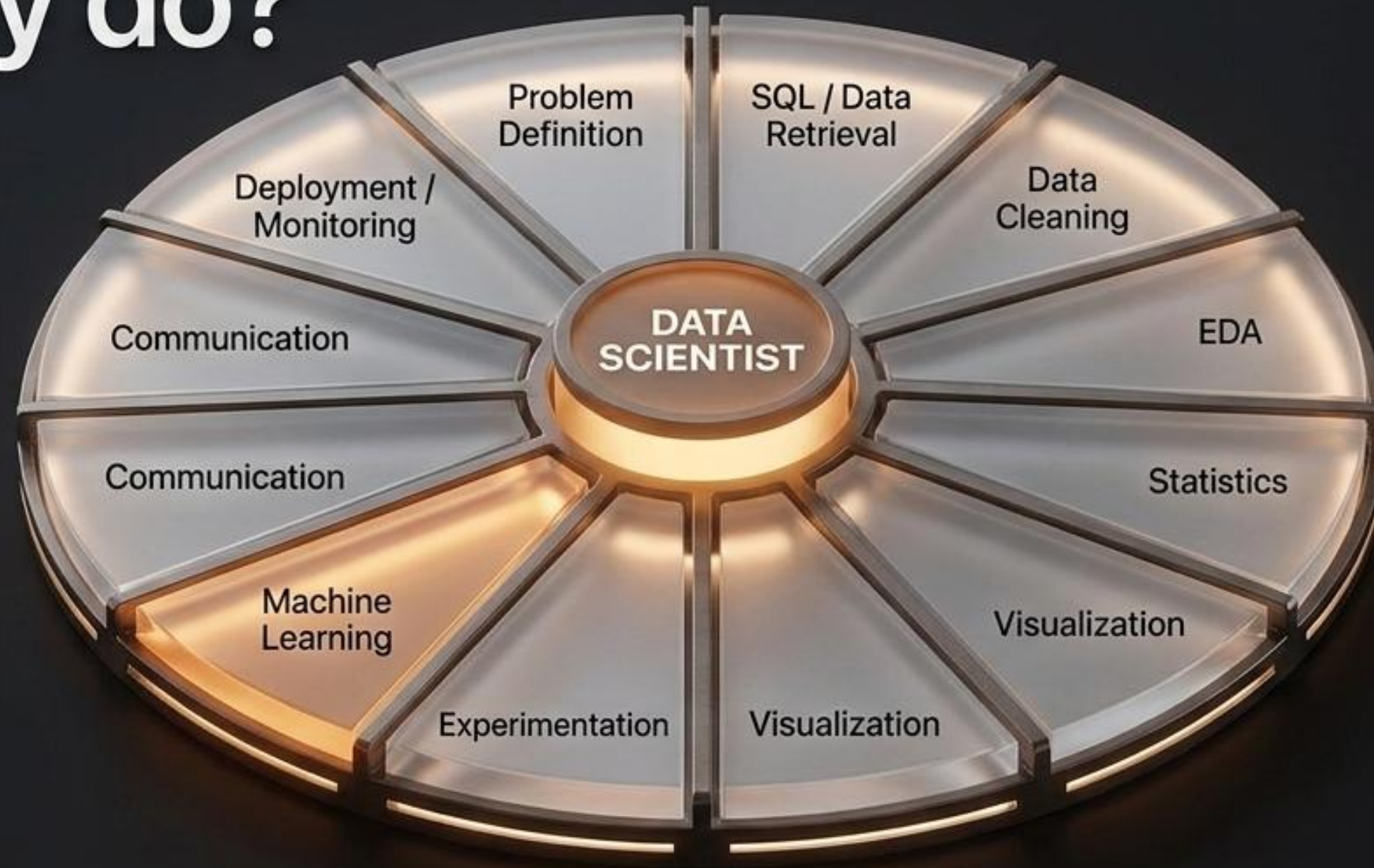
Right Side (The Real World)

Let's follow one complete Data Science problem

Illustrative Example: Subscription Churn



So... what does a Data Scientist actually do?

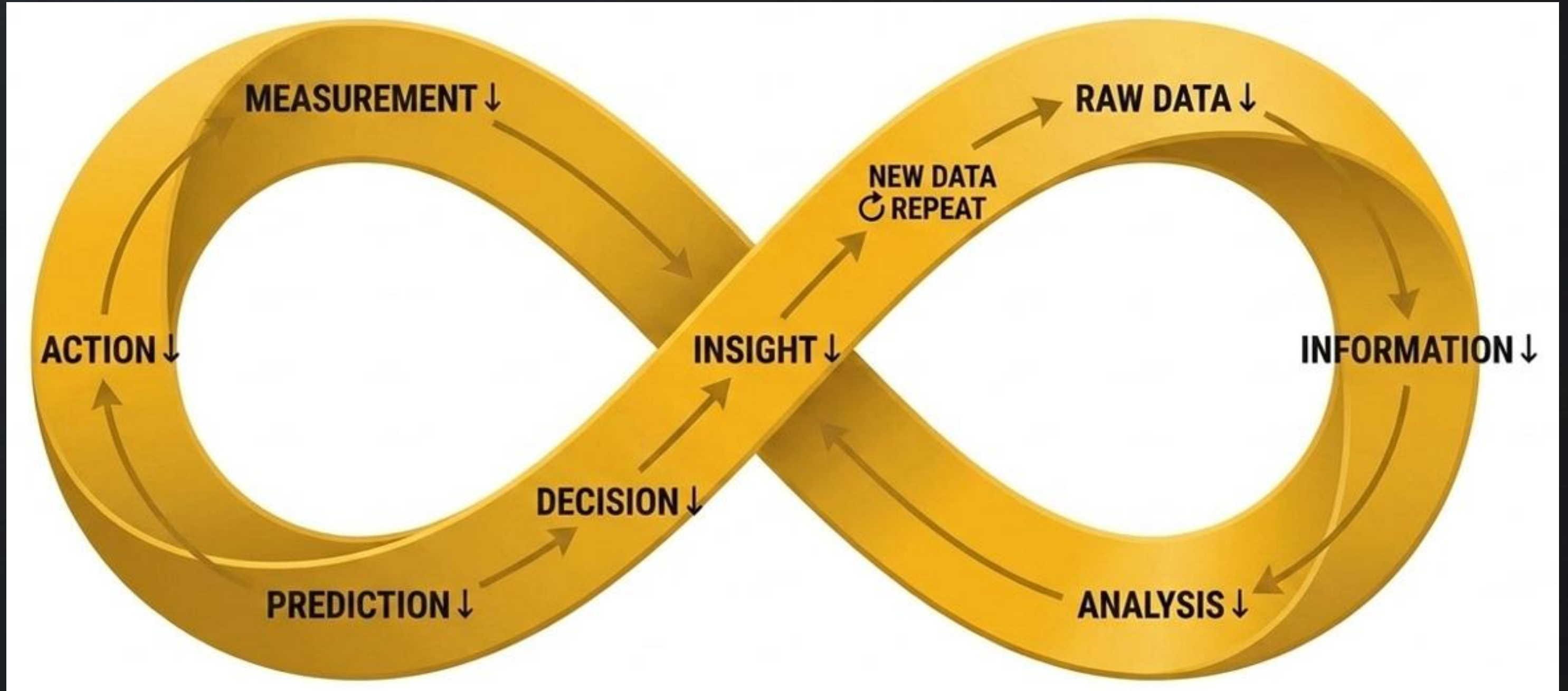


The job is solving problems with data—not just training models.

The 2026 Skill Stack



Data Science is a loop, not a destination



The goal isn't the most complicated model. The goal is better decisions and measurable impact.



THANK YOU!

The future belongs to those who **build it continuously.**



START TODAY.
Take the first step.



STAY CONSISTENT.
Small steps, big results.



BUILD ONE PROJECT
at a time.



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FOUNDER, THINKHUB



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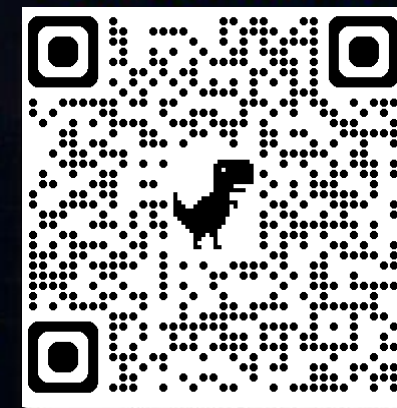


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